

Pablo Fernando Badilla Torrealba

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PROFESSIONAL SUMMARY

Computer Engineer and Master of Computer Science specialized in Data Science, Machine Learning, and Generative AI, with over 7 years of experience leading and implementing production systems for banking, retail, academia, and start-ups, among others.

Has led technical teams with a focus on client management, scalable system architecture, and mentoring, promoting the creation of high-standard products, best practices, and solid end-to-end execution.

WORK EXPERIENCE

Iniciativa de Datos e Inteligencia Artificial (IDIA) – Universidad de Chile

Santiago, Chile

Tech Lead, Technology Transfer Area

April 2026 – Present

- **Area Building & Definition:** Leadership in the definition and structuring of the technology transfer area within IDIA, an initiative that brings together the main academic departments of FCFM (DIE, DII, DCC, DIM) to articulate research, transfer, teaching, and project capabilities.
- **AI-based ETL – Ministerio de las Culturas:** Development of an automated indicator data extraction and transformation system (AI-based ETL) using Python, LLMs/Generative AI, and FastAPI, enabling the Ministry of Culture to process cultural indicator data automatically.

Banco Falabella

Santiago, Chile

Data Scientist

August 2024 – April 2026

- **Products & Benefits:** AI Agent-based chat for queries about products, benefits, operations, and branches in Chile, Colombia, Mexico, and Peru. 100k+ monthly interactions, 80% cost reduction, improved latency and user experience. Internal client relationship management across all countries in collaboration with the Digital Channels team.
- **Transaction Dispute:** Development and launch of a project focused on initiating a fraud dispute process for unreconciled transactions. First AI initiative published by the bank, opening the door to more complex projects and an AI-driven strategy. Collaboration with the Digital Channels team.
- **Concurrent Conversation Server Library:** Design and implementation of a library based on concurrent and distributed algorithms to coordinate conversations across different AI use cases. Foundation for new conversational AI projects.
- **Mentoring:** Mentoring on conceptual development of AI conversational agents, technologies (LangChain / LangGraph / LangFuse, RAG, etc.), backend technologies (FastAPI, SQLAlchemy, Pydantic, Asyncio, etc.), and AI-assisted development (Copilot) for the team.

Walmart Chile

Santiago, Chile

Machine Learning Engineer

April 2023 – April 2024

- **Call Center Timeseries:** Development and deployment of a Machine Learning timeseries model for daily demand forecasting at the Call Center. Error reduced from 60% to 20%. Evaluated via A/B testing and shadow deployment with ML metrics and cost-related KPIs. Milestone planning and joint presentation of progress and results to internal operations team clients together with the PO.
- **Queue Wait-Time Simulation:** Optimization of a project measuring queue wait times by store format (Lider, Express, A cuenta, Central Mayorista) and checkout type (MM1 and MMS) using queuing theory. Substantial architectural and performance refactoring of the system.
- **Customer Lifetime Value (CLV):** Reimplementation of the Customer Lifetime Value model for the Marketing area, enabling personalized campaigns by detected customer segment.

DashAI
Tech Lead

Santiago, Chile
January 2023 – June 2024

- **Development Team Leadership:** Technical leadership of a team of 5 developers (Universidad de Chile thesis students). Responsible for guiding various projects and initiatives on technical and architectural aspects.
- **Management & Coordination:** Team direction, ceremony facilitation (dailies, plannings, refinements, etc.), definition of technical and integration requirements. Repository: github.com/DashAISoftware/DashAI
- **Funding Application & Technology Transfer:** FONDEF 2023 grant application. Business model generation and plan to productize the application and its conceptual framework. Technology transfer strategy design, competitor analysis, and business model development.
- **Core Component Development:** Creation of Component Registry (Plugins), dependency injection, subproject integration, mass testing, initial front-end, and development best practices (CI, tests, static code verification, typing), etc.

BCI
Data Scientist

Santiago, Chile
February 2023 – May 2023

- **Credit Risk Assessment Features:** Creation of features based on customer transactional behavior to explore improvements in credit risk assessment models.

National Center for Artificial Intelligence – CENIA
Machine Learning Engineer

Santiago, Chile
January 2022 – January 2023

- **End-to-End ML Model for VRP:** Design and deployment of a Machine Learning model that incorporates user preferences into solutions for the Vehicle Routing Problem (VRP). Client communication, progress presentations, milestone and delivery management in collaboration with the Optimization team. Delivery of a complete package ready for deployment in internal systems.

Universidad de Chile
Part-Time Lecturer

Santiago, Chile
March 2021 – July 2024

- **Scientific Programming Laboratory for Data Science:** Lecturer for the Scientific Programming Laboratory for Data Science course in the Master in Data Science (MDS) program at Universidad de Chile.
- **Data Handling and Extraction in Python:** Lecturer for the Data Handling and Extraction in Python course in the Python Applied to Data Science Diploma at the Department of Computer Science, Universidad de Chile.

Kwiyx SPA
Full Stack Developer

Santiago, Chile
April 2020 – January 2021

- **Real Estate Investment App:** Front-end development of a real estate investment application.
- **Hospital HR Management:** Full-stack development of an application for human resources management in pandemic hospitals based on shift optimization.

Freelance Developer
Data Scientist & Full Stack Developer

Santiago, Chile
July 2019 – January 2021

- **Topic Detection System – Universidad de Chile & Fundación Tribu:** Construction of a topic detection system using Transformer embeddings to analyze the 2019 participatory process of “Lxs 400” around the new Chilean Constitution.
- **Automated Text Detection – Sotta y CIA:** Development of an OCR + Regex-based system for automatic text extraction from automotive credit promissory notes.

Acción Circular
Full Stack Developer

Santiago, Chile
April 2019 – July 2019

- **Recyclable Waste Management System:** Design and implementation of software for the management of recyclable waste collection.

Cognus Consulting

Data Analyst Intern

Santiago, Chile

December 2017 – February 2018

- **Data Warehousing & Business Intelligence Solution:** Construction of a Data Warehousing, Business Intelligence, and visualization solution using Pentaho.

PROJECTS

Word Embeddings Fairness Evaluation Framework (January 2020 – June 2025):

Leadership and development of a library for measuring and mitigating gender, ethnic, and religious biases in word embedding models (precursors to language models). 2 published papers, 182 GitHub stars.

Repository: github.com/dccuchile/wefe Documentation: wefe.readthedocs.io

EDUCATION

Master of Computer Science

Universidad de Chile

Santiago, Chile

2019 – 2020

Computer Engineering (Civil)

Universidad de Chile

Santiago, Chile

2013 – 2019

PUBLICATIONS

WEFE: A Python Library for Measuring and Mitigating Bias in Word Embeddings

Pablo Badilla, Felipe Bravo-Marquez, María José Zambrano, Jorge Pérez.

Journal of Machine Learning Research, Vol. 26, No. 156, pp. 1–6, 2025.

<http://jmlr.org/papers/v26/22-1133.html>

WEFE: The Word Embeddings Fairness Evaluation Framework

Pablo Badilla, Felipe Bravo-Marquez, Jorge Pérez.

Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence (IJCAI-20), pp. 430–436, 2020.

<https://doi.org/10.24963/ijcai.2020/60>

SKILLS & TOOLS

- **Languages:** Spanish (Native), English B2 TOEFL-ITP
- **Programming Languages:** Python (Advanced), JavaScript (Intermediate)
- **Generative AI:** LangChain, LangGraph, LangFuse, OpenAI API, Gemini, Azure
- **Data Science:** NumPy, Pandas, SQL, BigQuery, Spark, JupyterLab, Plotly, Databricks
- **Machine Learning:** Scikit-Learn, XGBoost, LightGBM, PyTorch, HuggingFace
- **Backend Development:** FastAPI, SQLAlchemy, Pydantic, Asyncio, Concurrency, Dependency Injection
- **Cloud & Infrastructure:** Google Cloud Platform, Vertex AI, Docker, Kubeflow
- **Tools & Methodologies:** Kedro, GitHub Actions, GitLab CI, GitHub Copilot, MLOps, Scrum
- **Operating Systems & Office:** Windows, Linux, macOS, Microsoft Office, Google Workspace, SharePoint, OneDrive

ADDITIONAL INFORMATION

- Graduated from Engineering and Master's with Highest Distinction.
- Outstanding Student 2016 and 2018: Faculty of Physical and Mathematical Sciences, Universidad de Chile.

- Thesis Supervisor: Implementation of a bias mitigation module for word embedding models in WEFE.
- Guest Lecturer: Talks on Machine Learning Operations for various undergraduate and graduate courses at Universidad de Chile.
- **Microsoft College Code Competition:** Third place in the Microsoft College Code Competition 2018.
- **Paper Reviewer:** Secondary reviewer at The Joint Conference of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL-IJCNLP 2021).